TryHackMe Writeup 6: Networking Concepts

Room: OSI and TCP/IP, IP addresses and subnets, TCP/UDP, encapsulation, routing, Telnet

Objective: Go deeper into the OSI model, IP addresses, subnetting, routing, TCP, UDP, encapsulation, and direct connections to TCP services.

Task 1 - Introduction

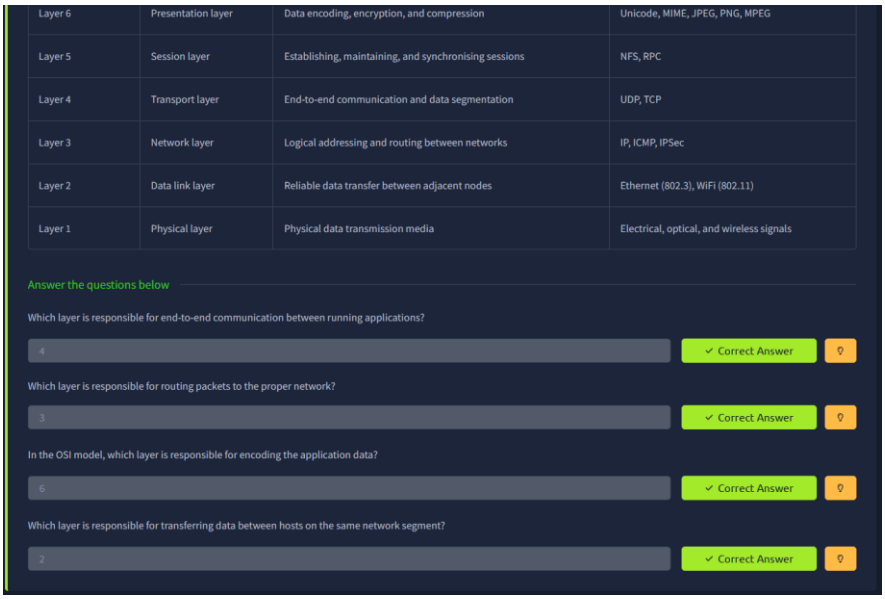
Read the introduction and review the room's learning objectives.

Learning point: A packet's journey depends on addressing, routing, transport protocols, and the link technology used at each network segment.

Task 2 - OSI Model

The OSI model is a conceptual framework with seven layers. It helps explain how data moves through a network and where a protocol or problem fits.

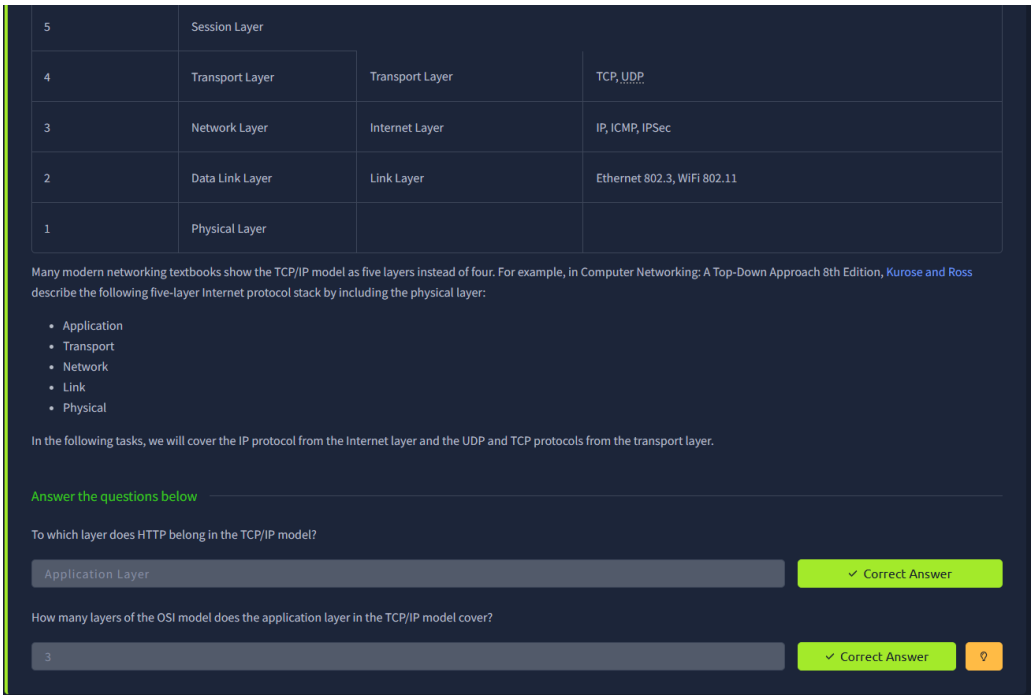
Layer	Name	Common responsibilities
7	Application	Network services used by applications
6	Presentation	Data representation, encoding, compression, and encryption-related functions
5	Session	Communication session management
4	Transport	End-to-end transport using TCP or UDP
3	Network	Logical addressing and routing
2	Data Link	Local network frames and link-layer delivery
1	Physical	Signals, cables, fiber, and radio transmission



Task 3 - TCP/IP Model

The commonly taught TCP/IP model has four layers: Application, Transport, Internet, and Network Access/Link. It groups some functions differently from the seven-layer OSI model.

The TCP/IP Application layer groups functions commonly associated with the OSI Application, Presentation, and Session layers.



5	Session Layer		
4	Transport Layer	Transport Layer	TCP, UDP
3	Network Layer	Internet Layer	IP, ICMP, IPSec
2	Data Link Layer	Link Layer	Ethernet 802.3, WiFi 802.11
1	Physical Layer		

Many modern networking textbooks show the TCP/IP model as five layers instead of four. For example, in Computer Networking: A Top-Down Approach 8th Edition, Kurose and Ross describe the following five-layer Internet protocol stack by including the physical layer:

- Application
- Transport
- Network
- Link
- Physical

In the following tasks, we will cover the IP protocol from the Internet layer and the UDP and TCP protocols from the transport layer.

Answer the questions below

To which layer does HTTP belong in the TCP/IP model?

Application Layer ✓ Correct Answer

How many layers of the OSI model does the application layer in the TCP/IP model cover?

3 ✓ Correct Answer

Task 4 - IP Addresses and Subnets

An **IP address** is a logical address used to identify an interface and route packets between networks. IPv4 addresses use 32 bits, commonly written as four decimal octets. A **subnet** is a defined portion of an IP network.

Private IPv4 ranges

- 10.0.0.0/8
- 172.16.0.0/12 (172.16.0.0–172.31.255.255)
- 192.168.0.0/16

Addresses outside these private ranges are not automatically guaranteed to be publicly routable; other special-use ranges exist.



Task 5 - TCP and UDP

TCP (Transmission Control Protocol) is connection-oriented and provides reliable, ordered delivery. It establishes a connection using a three-way handshake.

UDP (User Datagram Protocol) is connectionless and has lower protocol overhead, but it does not guarantee delivery or ordering.

The transport port-number range is 0–65535, giving 65,536 possible numeric values. Some values are reserved or have special conventions; not every port is necessarily available for every use.

TCP three-way handshake

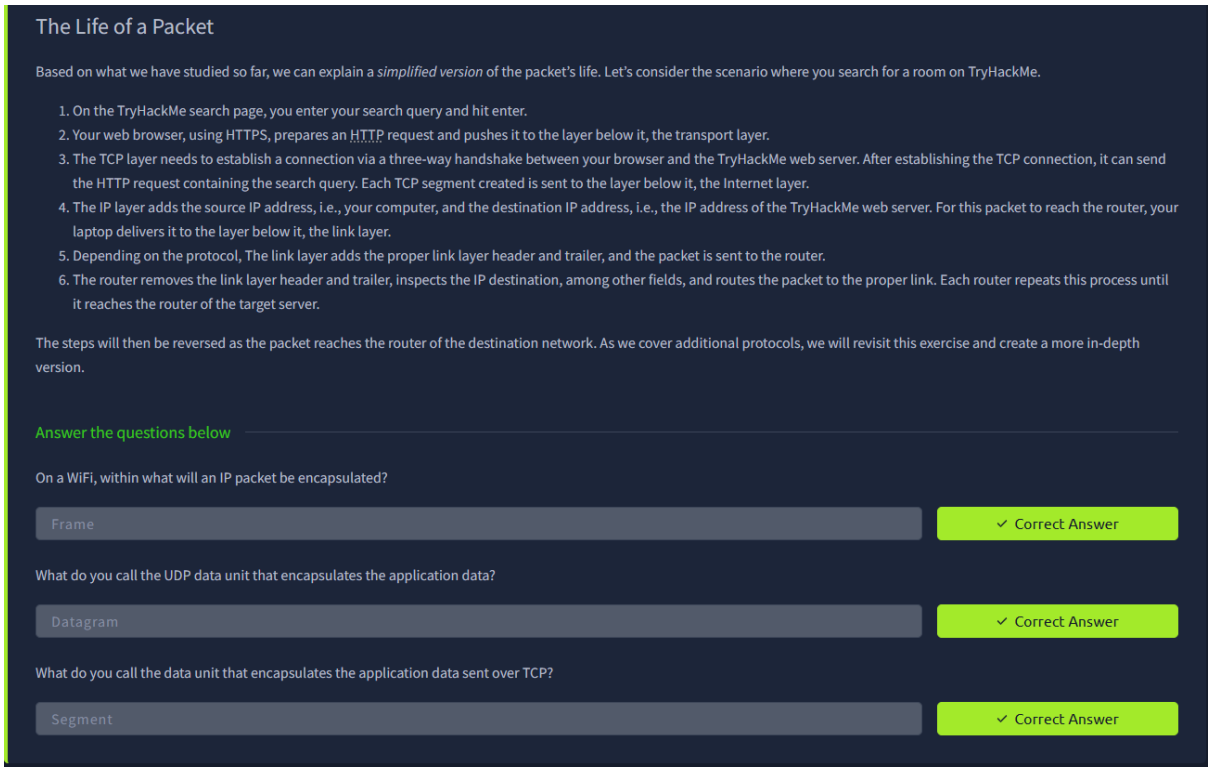
1. **SYN:** Client requests a connection.
2. **SYN-ACK:** Server acknowledges and responds.
3. **ACK:** Client acknowledges the response.



Task 6 - Encapsulation

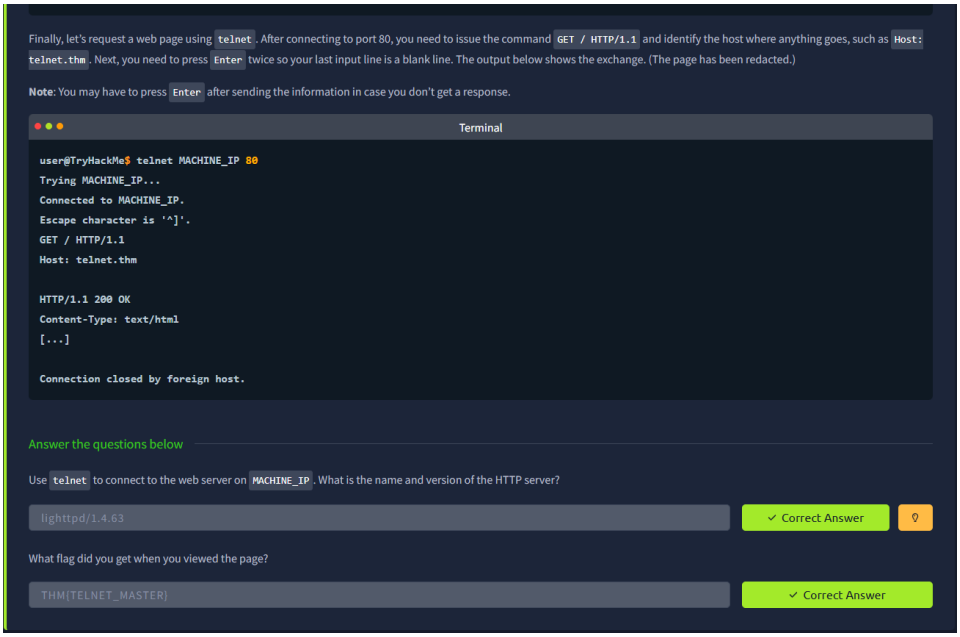
Encapsulation is the process of adding protocol information as data moves down the network stack. At the receiving end, de-encapsulation removes and processes that information.

Encapsulation organizes communication; by itself, it does not encrypt data or guarantee security.



Task 7 - Telnet

Telnet is a text-based protocol that can open a TCP connection to a specified host and port. Traditional Telnet does not encrypt the session, so credentials and data may be exposed. In this room, it is used to interact with a web server in the designated lab.



Practical steps

1. Start the TryHackMe lab machine and AttackBox/VPN environment as instructed.
2. Obtain the lab's MACHINE_IP from the room.

3. Connect to the target's HTTP port:

telnet MACHINE_IP 80

4. At the prompt, enter the HTTP request lines:
5. GET / HTTP/1.1

Host: telnet.thm

6. Press Enter twice so that the request ends with a blank line.
7. Inspect the HTTP response and page content displayed by the authorized lab.
8. Record the HTTP server name/version and flag shown by your session.

Security lesson

Telnet is useful for demonstrating how a text protocol communicates over TCP, but it should not be used for sensitive remote administration over untrusted networks because it lacks encryption. Use SSH for secure remote shell access and HTTPS for protected web traffic.

Task 8 - Conclusion

Review the room and mark it complete once all tasks are done.

Key learning outcomes

- The OSI model has seven layers; TCP/IP commonly uses four.
- IP addresses and subnets help identify network interfaces and organize routing.
- TCP provides connection-oriented, reliable delivery; UDP provides connectionless datagrams.
- Encapsulation adds headers (and, at the data-link layer, a trailer in common frame formats) as data moves through the stack.
- Telnet can demonstrate direct TCP interaction, but it does not protect session contents with encryption.
- Service banners can reveal software and version information; administrators should minimize unnecessary information exposure and keep services updated.